

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 3
Total Marks:30

Duration : 45 Minutes

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

1)

Solution Using Backtracking Search

Step 1: Assign D1

Possible shifts for D1:

- Morning
- Afternoon
- Night (Violates Constraint 1)

Choose the first valid option.

Assignment:

D1 → Morning

Constraint Check:

- D1 is not assigned Night

Step 2: Assign D2

Remaining shifts:

- Afternoon
- Night

Choose the first available shift.

Assignment:

D2 → Afternoon

Constraint Check:

- Only one doctor per shift

Step 3: Assign D3

Remaining shift:

- Night

Constraint Checks:

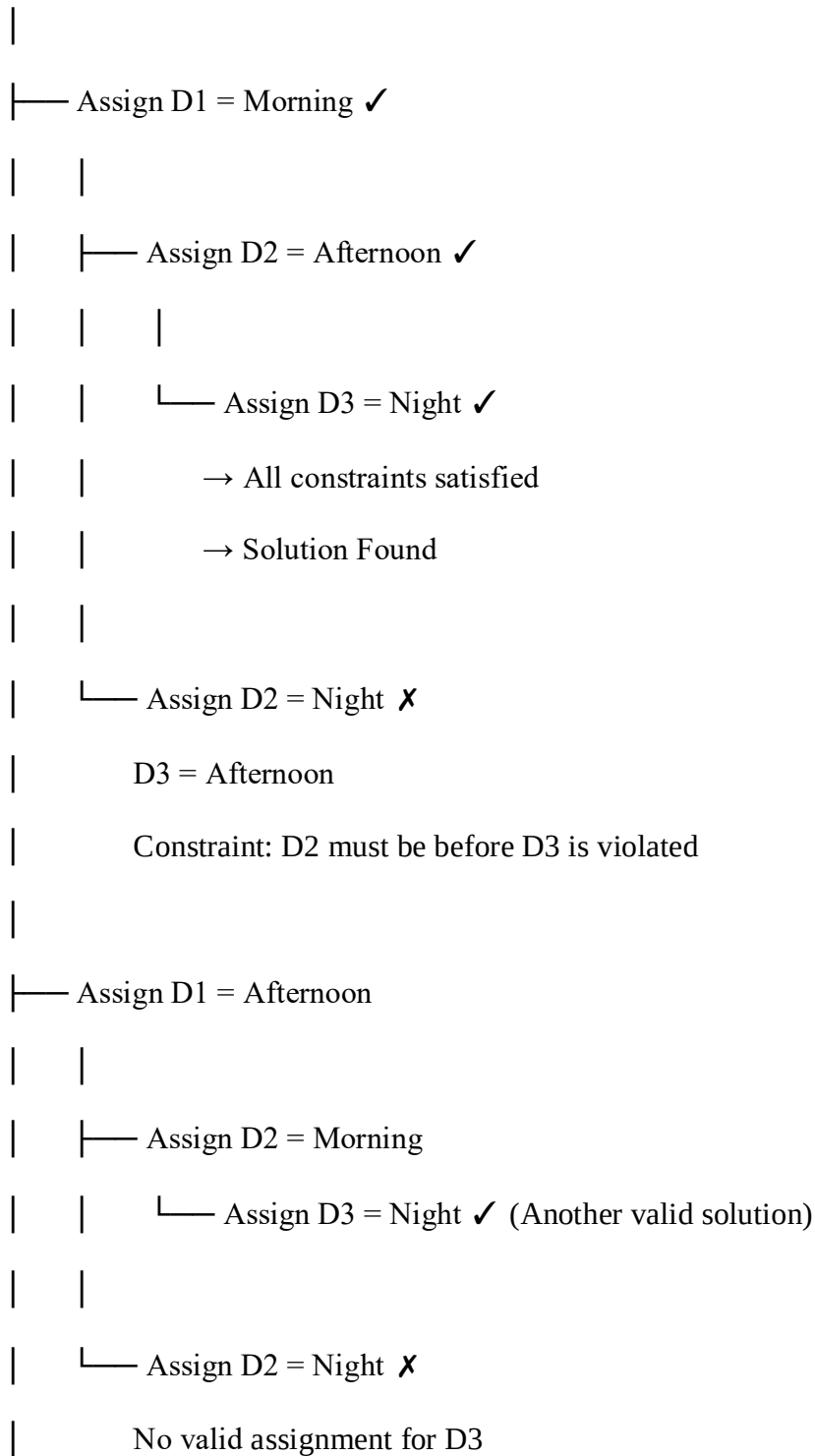
- D3 is not assigned Morning
- D2 works before D3:
 - D2 = Afternoon
 - D3 = Night
 - Afternoon < Night

- One doctor per shift
- Every doctor assigned exactly one shift

Since all constraints are satisfied, a valid solution is found.

Backtracking Search Tree

Start



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└─ Assign D1 = Night ✗

Violates Constraint 1

Intermediate Constraint Checks

Step	Assignment	Constraint Checked	Status
1	D1 → Morning	D1 cannot work Night	✓ Pass
2	D2 → Afternoon	Only one doctor per shift	✓ Pass
3	D3 → Night	D3 cannot work Morning	✓ Pass
4	D2 before D3	Afternoon < Night	✓ Pass
5	Final Assignment	Every doctor assigned exactly one shift	✓ Pass

Final Valid Schedule

Shift	Assigned Doctor
Morning	D1
Afternoon	D2
Night	D3

Conclusion

Using the Backtracking Search algorithm, invalid assignments were rejected whenever a constraint was violated, and the search continued until a valid assignment was obtained. The final schedule satisfies all five constraints.

Final Answer:

- Morning → D1
- Afternoon → D2
- Night → D3

2)

Constraints

1. Movement allowed: Up, Down, Left, Right
2. Cost of each move = 1
3. Robot cannot move through obstacles.

Grid Representation

C1 C2 C3 C4 C5

R1 S . X . .

R2 . . X . .

R3 . . . X .

R4 X

R5 . . X . G

Legend:

- S = Start
- G = Goal
- X = Obstacle
- . = Free Cell

Start = (R1,C1)

Goal = (R5,C5)

BFS Search Process

Step 1

Queue = [(R1,C1)]

Expand:

- (R1,C2)
- (R2,C1)

Queue after expansion:

[(R1,C2), (R2,C1)]

Step 2

Expand (R1,C2)

New node:

- (R2,C2)

Queue:

[(R2,C1), (R2,C2)]

Step 3

Expand (R2,C1)

New node:

- (R3,C1)

Queue:

[(R2,C2), (R3,C1)]

Step 4

Expand (R2,C2)

New node:

- (R3,C2)

Queue:

[(R3,C1), (R3,C2)]

Step 5

Expand (R3,C1)

No new valid node.

Queue:

[(R3,C2)]

Step 6

Expand (R3,C2)

New node:

- (R3,C3)
- (R4,C2)

Queue:

[(R3,C3), (R4,C2)]

Step 7

Expand (R3,C3)

New node:

- (R4,C3)

Queue:

[(R4,C2), (R4,C3)]

Step 8

Expand (R4,C2)

New node:

- (R5,C2)

Queue:

[(R4,C3), (R5,C2)]

Step 9

Expand (R4,C3)

New node:

- (R4,C4)

Queue:

[(R5,C2), (R4,C4)]

Step 10

Expand (R5,C2)

No new valid node.

Queue:

[(R4,C4)]

Step 11

Expand (R4,C4)

New nodes:

- (R4,C5)
- (R5,C4)

Queue:

[(R4,C5), (R5,C4)]

Step 12

Expand (R4,C5)

New node:

- (R5,C5) Goal Found

Queue:

[(R5,C4), (R5,C5)]

Search Stops.

Node Expansion Table

Step	Expanded Node	Queue After Expansion
1	(R1,C1)	[(R1,C2),(R2,C1)]
2	(R1,C2)	[(R2,C1),(R2,C2)]
3	(R2,C1)	[(R2,C2),(R3,C1)]
4	(R2,C2)	[(R3,C1),(R3,C2)]
5	(R3,C1)	[(R3,C2)]
6	(R3,C2)	[(R3,C3),(R4,C2)]
7	(R3,C3)	[(R4,C2),(R4,C3)]
8	(R4,C2)	[(R4,C3),(R5,C2)]
9	(R4,C3)	[(R5,C2),(R4,C4)]
10	(R5,C2)	[(R4,C4)]
11	(R4,C4)	[(R4,C5),(R5,C4)]
12	(R4,C5)	Goal Found

Optimal Path

(R1,C1)

→ (R2,C1)

→ (R3,C1)

→ (R3,C2)

→ (R3,C3)

→ (R4,C3)

→ (R4,C4)

→ (R4,C5)

→ (R5,C5)

Path Cost

- Total moves = 8
- Cost per move = 1

Total Cost = 8

Conclusion

The Breadth-First Search (BFS) algorithm explores the grid level by level using a FIFO queue, guaranteeing the shortest path in an unweighted grid. The robot successfully reaches the goal while avoiding obstacles.

Final Answer:

- **Optimal Path:** (R1,C1) → (R2,C1) → (R3,C1) → (R3,C2) → (R3,C3) → (R4,C3) → (R4,C4) → (R4,C5) → (R5,C5)
- **Total Cost:** 8

3)

Problem Statement

An autonomous rescue robot is deployed in a disaster-affected building to locate survivors. The robot starts at **S** and must reach the survivor at **G** while avoiding obstacles and minimizing the total movement cost.

(a) Constraint Analysis

Constraint	Effect on Robot's Decision Making
Move only Up, Down, Left, Right	Restricts movement to four directions only.

Constraint	Effect on Robot's Decision Making
Obstacles cannot be crossed	Robot must detect blocked cells and search for alternate paths.
Limited sensing (adjacent cells only)	Robot cannot view the entire map initially and must explore gradually.
Movement cost = 1	Every movement increases the total path cost.
Risky zone cost = +2	Robot avoids risky areas unless they produce a lower overall cost.
Dynamic knowledge update	Robot continuously updates its map as new cells are discovered.
Goal is to minimize total cost	Robot selects the least-cost safe path instead of simply the shortest path.

(b) Solution Using Uniform Cost Search (UCS) with Online Search

Step 1: Initialization

- Robot starts at S.
- Known information contains only the current cell and adjacent cells.
- Priority Queue is initialized with the start node having Cost = 0.

Step 2: Sense Environment

- Observe adjacent cells.
- Identify:
 - Free cells
 - Obstacles
 - Risky zones

Update the internal map.

Step 3: Expand Lowest-Cost Node

- Remove the node with the minimum cumulative cost from the priority queue.
- Generate all valid neighbouring cells.

Step 4: Calculate Cost

For every neighbouring cell:

- Normal cell:

New Cost = Previous Cost + 1

- Risky cell:

New Cost = Previous Cost + 3
(1 movement + 2 risk cost)

Store nodes in the priority queue according to the minimum cumulative cost.

Step 5: Dynamic Update

If a newly discovered obstacle blocks the current planned path:

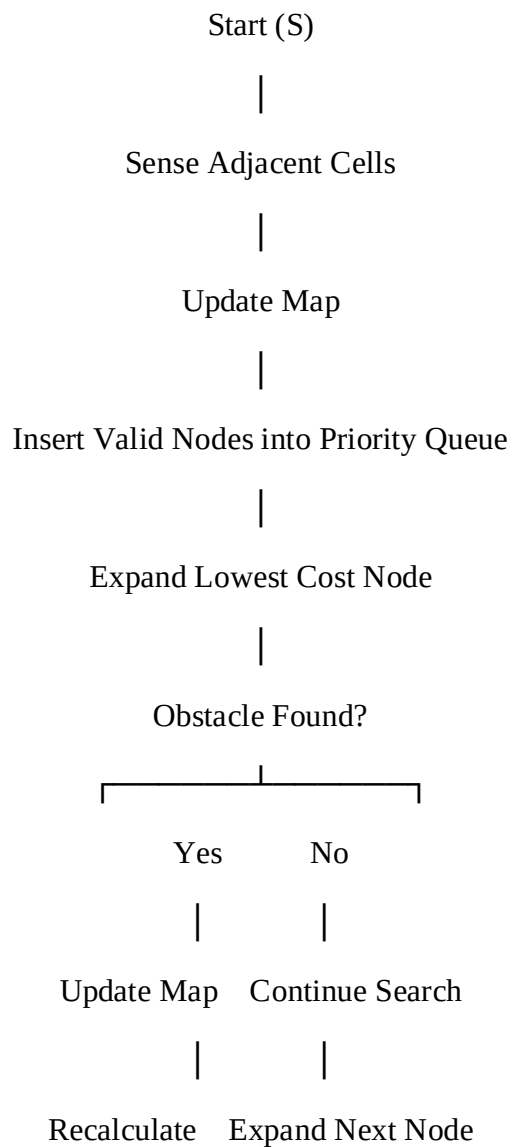
- Update the map.
- Remove invalid paths.
- Continue searching using the updated information.

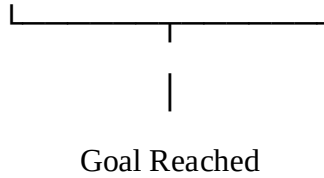
Step 6: Goal Test

When the robot reaches G, stop the search.

The selected path is the minimum-cost path discovered.

Working Flow





(c) AI Concepts Applied

1. Intelligent Agent

The rescue robot acts as an Intelligent Agent because it:

- Perceives the environment using sensors.
- Processes the available information.
- Makes decisions autonomously.
- Performs actions using actuators.

2. Rationality

The robot is rational because it:

- Chooses the action with the lowest total cost.
- Avoids obstacles.
- Avoids risky zones whenever possible.
- Continuously updates its decisions using new information.

3. Environment Type

Property	Description
Partially Observable	Robot can observe only adjacent cells.
Dynamic	Environment knowledge changes during movement.
Sequential	Current action affects future decisions.
Discrete	Grid consists of separate cells.
Single Agent	Only one rescue robot performs the task.

(d) Justification of the Chosen Approach

Uniform Cost Search (UCS) combined with an Online Search Agent is the most suitable approach because:

1. It always expands the node with the lowest cumulative cost.
2. It considers both movement cost and risk cost.
3. It guarantees the optimal least-cost path when costs are non-negative.
4. It works effectively in a partially observable environment by updating information during execution.
5. It adapts immediately when new obstacles or risky zones are discovered.
6. It provides safe and efficient navigation for rescue operations.

Conclusion

The rescue robot uses Uniform Cost Search (UCS) with online search to navigate safely in a partially observable disaster environment. By continuously sensing nearby cells, updating its knowledge, avoiding obstacles, accounting for risky zones, and selecting the lowest-cost path, the robot behaves as a rational intelligent agent capable of reaching the survivor efficiently.

Chosen Search Strategy: Uniform Cost Search (UCS) with Online Search Agent.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25 %	Clearly defines the problem and identifies all relevant constraints accurately.	Identifies most constraints with minor gaps.	Partial understanding; misses key constraints.	Fails to identify constraints correctly.
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically.	Provides reasonable analysis with some prioritization.	Limited analysis; weak prioritization.	No meaningful analysis or prioritization.
Solution Development	25 %	Develops optimal and feasible solutions satisfying all constraints.	Develops workable solutions with minor limitations.	Solutions partially satisfy constraints.	Solutions do not meet constraints.
Application of Concepts	15 %	Effectively applies appropriate concepts, methods, or tools.	Applies concepts with minor errors.	Limited application; requires guidance.	Unable to apply relevant concepts.
Justification	10 %	Provides strong justification for decisions with logical reasoning.	Provides justification with some clarity.	Weak or unclear justification.	No justification provided.